Research Project Report

Investigation on the Impact and Predictive Power of Short-Term Delayed Air Pollution on Hospital Admissions for Cardiovascular Problems

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Abstract

This report investigates the relationship between delayed air pollution and hospital admissions in a cardiology unit through statistical analysis and explores its predictive potential. After a quick exploratory analysis, we determine the optimal lag for each air quality variable by analyzing their delayed correlations with hospital admissions. Using these lagged variables, we perform a Linear Regression Analysis to assess their impact, identifying statistically significant predictors. We then build a binary classification model using Logistic Regression to predict days with higher-than-average hospital admissions, through an optimization-based feature selection process. Finally, we conduct a one-way ANOVA to examine seasonal variations in air quality, particularly on the Air Quality Index (AQI), shedding light on environmental patterns that may influence air quality and health outcomes.

1 Introduction

Understanding the factors that influence hospital admissions is critical for healthcare resource allocation and public health planning. Cardiovascular diseases (CVDs) remain one of the leading causes of morbidity and mortality worldwide [1], with hospital admissions imposing a significant burden on healthcare systems. While many factors contribute to these admissions, environmental conditions, particularly air quality, have garnered attention for their potential impact on cardiovascular health.

Air pollution has been linked to adverse health outcomes, including exacerbations of respiratory and cardiovascular conditions. Short-term exposure to pollutants such as particulate matter ($PM_{2.5}$, PM_{10}), nitrogen dioxide (NO_2), sulfur dioxide (SO_2), and carbon monoxide (CO) is known to trigger acute cardiovascular events [2]. Hospitals and policymakers need to understand these short-term impacts to anticipate high-admission days and allocate resources efficiently.

However, the relationship between air quality and hospital admissions is complex. The impact of air pollution may be delayed, as the effects of exposure often manifest hours or days after initial contact. Moreover, each individual pollutant might exert its impact at a specific lag, as the physiological response to exposure can vary depending on the pollutant's properties, which complicates further our understanding of these dynamics [3].

Therefore, our study aims at (i) determining the most correlated lag for each air quality variable (ii) assessing the statistical significance of these lagged impacts through a Linear Regression Analysis (iii) developing a binary classification model to investigate their predictive potential (iv) examining seasonal variations in air quality through a one-way ANOVA.

2 Data Exploration

2.1 Dataset Creation and Description

The raw data used in this study span two years, from April 1, 2017, to March 31, 2019, was originally collected for this research [4] and then made publicly available at [5]. We will focus our work on two datasets: (1) *Admissions Dataset*, which contains dated records of all hospital admissions to the cardiology unit, totaling 14,845 entries over the study period, and (2) *Pollution Dataset*, which provides daily environmental and air quality metrics such as particulate matter ($PM_{2.5}, PM_{10}$), nitrogen dioxide (NO_2), sulfur dioxide (SO_2), carbon monoxide (CO), and meteorological data like temperature and humidity. The *Admissions Dataset* was aggregated by date to compute daily *Total Admissions*, then merged with the *Pollution Dataset* by date, resulting in a unified dataset of 725 rows (one per day) and 28 features. Throughout this report, hospital admissions specifically refer to those in the cardiology unit.

2.2 Exploratory Analysis

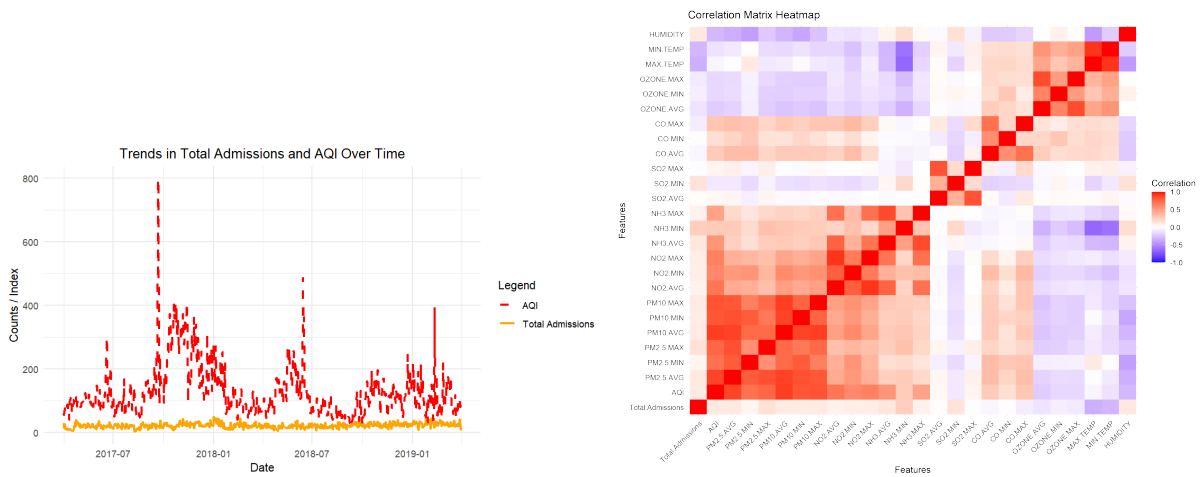

Figure 1: Plots of AQI [6] and Total Admissions Over Time (left), and Correlation Matrix Heatmap (right).

The exploratory analysis aims to provide an initial understanding of the dataset and to motivate the direction of our research. Time-series plots of hospital admissions and air quality indicators do not reveal an obvious pattern.

The correlation heatmap highlights limited Pearson correlations between the zero-lag features (current day) and the target variable, with the only notable exceptions being the minimum and maximum temperatures, which exhibit absolute correlations of approximately 0.3.

Since same-day air pollutant concentrations do not appear to be strongly correlated with hospital admissions, this motivates the exploration of lagged effects, as delayed impacts may provide more insights of the relationship between air quality and hospital admissions.

The target variable seems to have few outliers, but since there is one row per day, we decided to keep them because it would create time-dependency problems in the following sections.

(See Appendix A for the box plot of the target variable).

3 Linear Regression Analysis of Lagged Variables

To assess the delayed effects of air quality on hospital admissions, we performed a linear regression analysis using lagged variables.

3.1 Feature Engineering and Pre-Selection

To capture the delayed impact of pollutants, we generated 30 lagged variables for each air quality feature X_i, representing their values from the current day (lag 0) to 30 days prior (lag 30). Specifically, lagged variables $X_i^{(t-k)}$ were created, representing the value of feature X_i k days prior to day t, for all $k \in [0, 30]$. To identify the most relevant lag for each feature, the Pearson correlation coefficient $\rho_{X_i^{(t-k)},Y}$ [7] between each lagged variable and the target variable, *Total Admissions* Y was computed, and the lag with the highest absolute correlation was selected for each feature. Formally, the optimal lagged variable X_i^* was determined as:

$$X_i^* = X_i^{(t-k^*)}, \quad \text{where } k^* = \arg\max_{k \in [0,30]} |\rho_{X_i^{(t-k)},Y}|.$$

This pre-selection reduced the dataset's dimensionality while retaining the most relevant lagged variables. Since lagged variables require data from prior observations, the first 30 rows (corresponding to the first 30 days) were excluded. Note that we also tested the use of Lagged Rolling Averages to capture cumulative effects; however, this approach was found to reduce the variance of the predictors, limiting their explanatory power. The full list of pre-selected features with their corresponding lags and correlations with the target can be found in Appendix B, we can see a clear improvement compared to the correlation matrix in Figure 1.

3.2 Model Construction and Feature Selection

A stepwise regression approach was used to construct the final linear regression model. This method iteratively adds or removes predictors based on the Akaike Information Criterion (AIC), with a penalty term ($k = 4$) introduced to encourage the selection of a simpler model. The resulting model included 11 selected features with their corresponding lags. The full list of selected lagged features is in Appendix C, with the results discussed in Subsection 3.5.

3.3 Linear Regression Analysis with Selected Predictors

Using the predictors selected through the stepwise approach, we fitted a linear regression model [7] to examine the relationship between lagged air quality variables and hospital admissions. The regression model is expressed as:

$$Y = \beta_0 + \sum_{i=1}^{11} \beta_i X_i^* + \epsilon,$$

where Y represents daily hospital admissions, X_i^* are the selected predictors, β_i are the coefficients, and ϵ is the error term.

3.4 Assumptions Validation

To ensure the reliability of the model, we validated the assumptions of the linear regression model. **Independence of Residuals:** The Durbin-Watson test [8] yielded a statistic of 1.6, indicating mild autocorrelation but within an acceptable range of 2 ± 0.5. **Homoscedasticity:** Residual plots against fitted values showed no clear pattern, suggesting constant variance. **Normality of Residuals:** Residuals approximately followed a normal distribution, as confirmed by the histogram and Q-Q plot. See Appendix D for supporting outputs.

3.5 Results and Discussion

The complete R-output of our model is provided in Appendix C. The adjusted R^2 value of 0.2053 indicates that the model explains approximately 20.53% of the variance in hospital admissions.

The residual standard error of 6.729 admissions reflects the average deviation of observed values from predictions. The F-statistic of 17.3 (p-value $< 2.2 \times 10^{-16}$) confirms the overall significance of the model. All 11 selected predictors have p-values below 0.05, indicating statistically significant contributions and providing evidence that the feature pre-selection/selection was rigorous. Finally, we can interpret the coefficients' practical implications. For instance, the coefficient of 0.2586 for `SO2.MIN(Lag 28)` measured in $\mu g/m^3$ implies that every 10 $\mu g/m^3$ increase in minimum sulfur dioxide concentration, measured 28 days prior, corresponds to an approximate 2.59 increase in today's predicted hospital admissions. This represents about 12.0% of the mean admissions (21.58). While all predictors are statistically significant, the model's moderate explanatory power suggests that additional factors not captured in this study likely contribute to variability in admissions.

4 Predictive Power in a Binary Classification Problem

Following our previous work, this section is dedicated to assessing the predictive power of the lagged variables on hospital admissions through a practical binary classification problem. This approach was chosen for its simplicity in evaluation.

4.1 Target Transformation and Dataset Preparation

The target variable *Total Admissions* was binarized, assigning a value of 1 to days exceeding the 66th percentile ($p = \frac{2}{3}$), therefore considered as days with higher-than-usual admissions and 0 otherwise. The same pre-selection method from Subsection 3.1 was applied to our new dataset, generating the features X_i^*. See Appendix E for the full list of pre-selected features.

4.2 Feature Selection with a Greedy Algorithm

For a single observation, the logistic regression model [7] expresses the probability of the binary response $Y \in \{0, 1\}$ as:

$$P(Y = y_i \mid X^* = x_i^*) = \left(\frac{1}{1 + e^{-(\beta_0 + \mathbf{x}_i^* \cdot \boldsymbol{\beta})}} \right)^{y_i} \cdot \left(1 - \frac{1}{1 + e^{-(\beta_0 + \mathbf{x}_i^* \cdot \boldsymbol{\beta})}} \right)^{1 - y_i},$$

where $\mathbf{x}_i^*$ is the vector of pre-selected predictors for observation i, and $\boldsymbol{\beta}$ is the vector of their corresponding coefficients. For all observations, the likelihood function is constructed as:

$$\mathcal{L}(\beta \mid X^*, Y) = \prod_{i=1}^{n} P(Y = y_i \mid X^* = x_i^*),$$

where n is the total number of observations. Let S be a subset of predictors, X_S the matrix of predictors in S, and $\hat{\boldsymbol{\beta}}_S$ the maximum likelihood estimators of their coefficients. Maximizing $\mathcal{L}$ identifies the parameters that best explain the data, the feature selection (FS) problem is therefore formulated as:

$$S^* = \arg \max_{S \subseteq \{X_1^*, \ldots, X_{26}^*\}} \mathcal{L}(\hat{\boldsymbol{\beta}}_S \mid X_S, Y).$$

Due to the computational complexity of evaluating all 2^{26} subsets, a greedy algorithm [9] was used to find a suboptimal solution. The algorithm begins with an empty set $S = \emptyset$ and iteratively adds the predictor X_i^* that maximizes the marginal likelihood improvement:

$$X_i^* = \arg \max_{X_i \notin S} \left(L(\hat{\boldsymbol{\beta}}_{S \cup \{X_i\}} \mid X_{S \cup \{X_i\}}, Y) - L(\hat{\boldsymbol{\beta}}_S \mid X_S, Y) \right).$$

The process terminates when the improvement falls below a predefined threshold ($\epsilon = 1$), ensuring that only meaningful predictors are retained. This approach provides a computationally efficient FS method, avoiding overfitting while prioritizing predictors with the highest contribution. See Appendix F for the selected features and their marginal likelihood improvement.

4.3 Model Evaluation and Validation

All outputs from this Subsection are in Appendix F. We first note that all 8 over the 11 selected features are considered statistically significant (p-value $< 5\%$). To evaluate our model, a random split was chosen over a time-based split due to moderate autocorrelation (From Subsection 3.4 Durbin-Watson statistic= ~ 1.6) and potential seasonal biases (see Section 5) that could significantly affect the results. The dataset was divided into 80% training and 20% testing, yielding: **Accuracy**: 0.741, **ROC-AUC [10]** : 0.715, **F1 score**: 0.514, which achieves better performance than simply always predicting the majority class.

Additionally, the 5-fold cross-validation provided mean results of: **Accuracy**: 0.722, **ROC-AUC**: 0.737, **F1 score**: 0.506, with small variance observed across the folds. These results indicate consistent performance across folds, demonstrating the model's ability to generalize with satisfactory predictive power.

5 Seasonal Variance Analysis in Air Quality Index (AQI)

The goal of this section is to investigate whether the Air Quality Index (AQI) [6] varies significantly across seasons, using ANOVA. This variable is interesting to analyse since, even if it was not selected in the previous models, it serves as a summary for many other polluants. Seasons were defined on the basis of meteorological criteria. Since Levene's Test [11] revealed unequal variances in AQI across seasons ($p\text{-}value < 0.05$), we decided to use Welch's ANOVA [12], which doesn't require homogeneity of variance. The results showed significant seasonal differences in AQI ($p\text{-}value = 4.14 \times 10^{-16}$). Additionally, Appendix G illustrates the AQI distribution for each season, highlighting notable variability, with higher mean values and extreme values during Winter and Fall. These findings emphasize the importance of considering seasonal effects when analyzing air quality's impact, as different seasons may introduce biases or trends in the data, and even provide an argument for the choice of a random split in Subsection 4.3

6 Conclusion

This study provides valuable insights into the relationship between air pollution and hospital admissions for cardiovascular problems. Meaningful delayed effects of specific air quality indicators, such as sulfur dioxide and carbon monoxide, were identified, with lagged variables serving as meaningful predictors. The linear regression model established statistically significant relationships, while the binary classification model demonstrated moderate predictive performance (Accuracy: ~ 0.74, ROC-AUC: ~ 0.72, F1 score: ~ 0.51) and consistent results during the cross validaton. Finally, seasonal analysis revealed higher pollutant concentrations and variability in winter and fall, emphasizing temporal considerations in public health planning. These findings offer practical implications for healthcare resource allocation and environmental policies.

Despite its valuable insights, the study has several limitations: The analysis used data from a single hospital unit over two years, limiting its generalizability. The model achieved only moderate explanatory power (Adjusted R^2: 0.2053) and predictive performance, suggesting that other relevant factors, such as demographic data or comorbidities were not captured. The use of a greedy algorithm, while computationally efficient, may overlook synergistic effects. Then, the lagged variables, while insightful, do not account for cumulative pollution exposure.

Future work could address these gaps by incorporating richer datasets, including more factors and exploring non-linear modeling techniques such as Generalized Additive Models (GAMs) [12] or ensemble methods to better capture true relationships. Additionally, more advanced methods like Long Short-Term Memory (LSTM) networks [14] or distributed lag models (DLMs) [15] could evaluate both lagged and cumulative effects. Finally, expanding the analysis to multiple hospital departments or regions would further enhance the applicability of these findings.

References

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Appendix

A Appendix A: Distribution of target variable

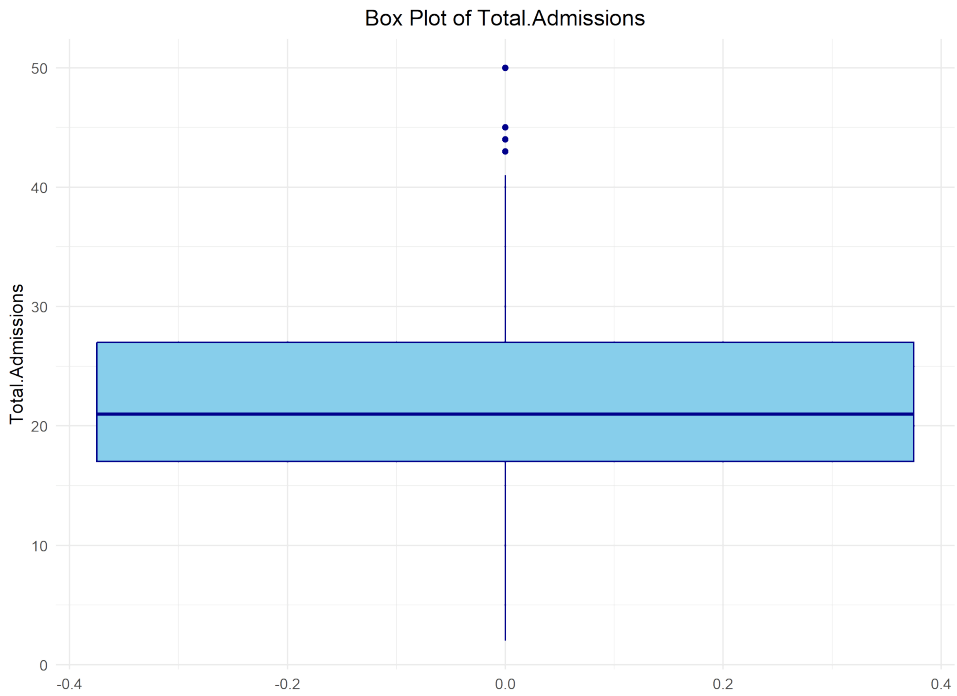


Figure 2: Box plot of the target variable.

B Appendix B: Pre-selection of Lagged Variables (LinReg)

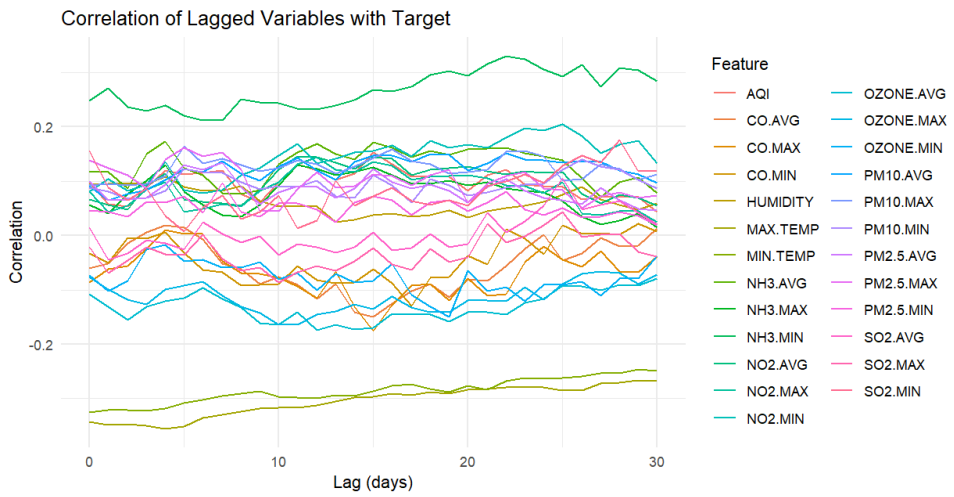


Figure 3: Line plot of the correlation of all lagged features with the target variable (LinReg).

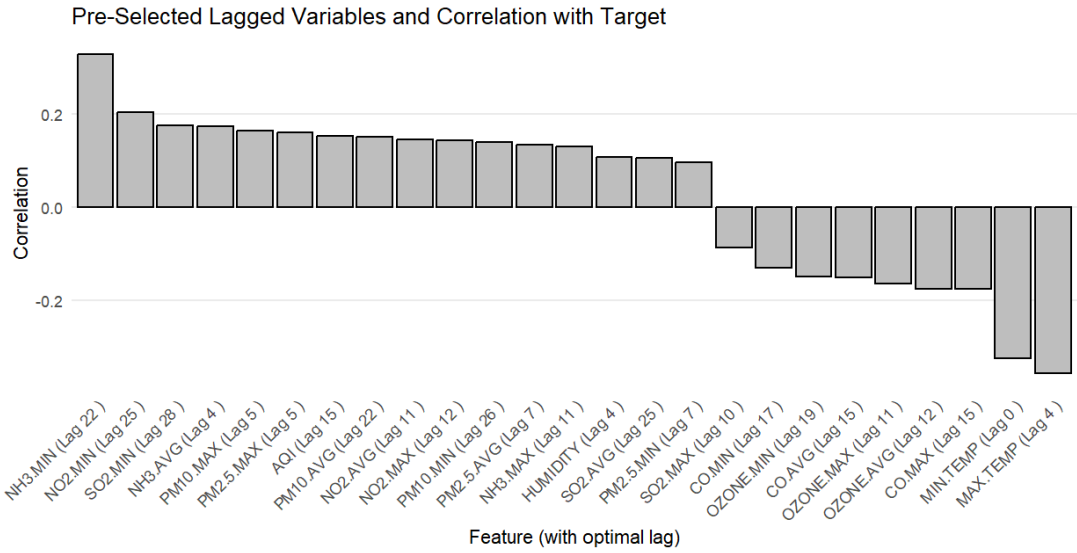


Figure 4: Bar plot of pre-selected lagged features with their correlation to the target variable (LinReg).

C Appendix C: Linear Regression Outputs

```
Call:
lm(formula = Total.Admissions ~ PM2.5.MAX_lag5 + NO2.AVG_lag11 +
    NO2.MAX_lag12 + NH3.MIN_lag22 + NH3.MAX_lag11 + SO2.AVG_lag25 +
    SO2.MIN_lag28 + SO2.MAX_lag10 + CO.MIN_lag17 + CO.MAX_lag15 +
    MAX.TEMP_lag4, data = final_data)

Residuals:
     Min       1Q   Median       3Q      Max
-18.4749  -4.2393  -0.2426   4.1591  20.8327

Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)    28.456360   1.965421  14.479  < 2e-16 ***
PM2.5.MAX_lag5  0.006950   0.002372   2.930  0.00350 **
NO2.AVG_lag11   0.023166   0.007431   3.117  0.00190 **
NO2.MAX_lag12   0.012208   0.004655   2.623  0.00891 **
NH3.MIN_lag22   0.165322   0.078517   2.106  0.03561 *
NH3.MAX_lag11  -0.067868   0.024613  -2.757  0.00598 **
SO2.AVG_lag25   0.083568   0.036533   2.287  0.02247 *
SO2.MIN_lag28   0.258600   0.100425   2.575  0.01023 *
SO2.MAX_lag10  -0.027396   0.012346  -2.219  0.02682 *
CO.MIN_lag17   -0.061552   0.026001  -2.367  0.01820 *
CO.MAX_lag15   -0.051287   0.010982  -4.670 3.63e-06 ***
MAX.TEMP_lag4  -0.201620   0.048549  -4.153 3.70e-05 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 6.729 on 683 degrees of freedom
Multiple R-squared:  0.2179,    Adjusted R-squared:  0.2053
F-statistic:  17.3 on 11 and 683 DF,  p-value: < 2.2e-16
```

Figure 5: Summary of Linear Regression Output.

D Appendix D: Residual Diagnostics

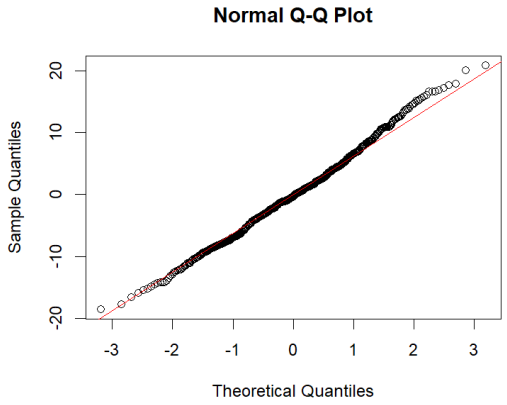


Figure 6: Normal Q-Q Plot.

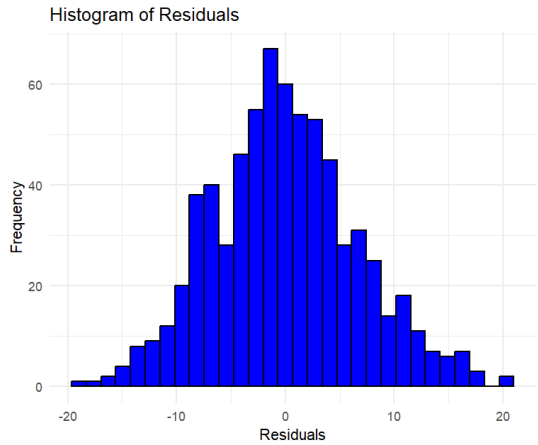


Figure 7: Histogram of Residuals.

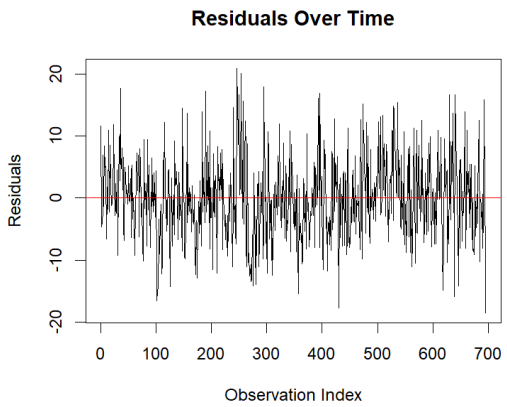


Figure 8: Residuals Over Time.

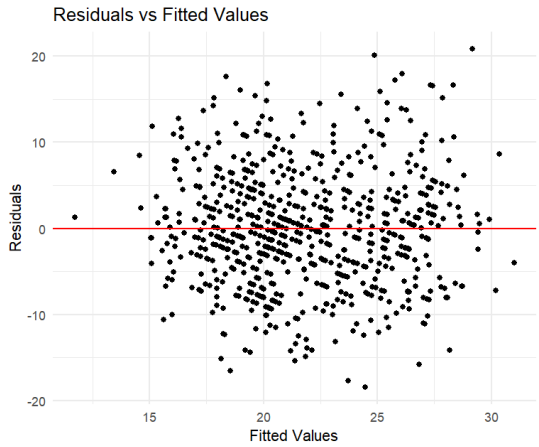


Figure 9: Residuals vs Fitted Values.

E Appendix E: Pre-selection of Lagged Variables (LogReg)

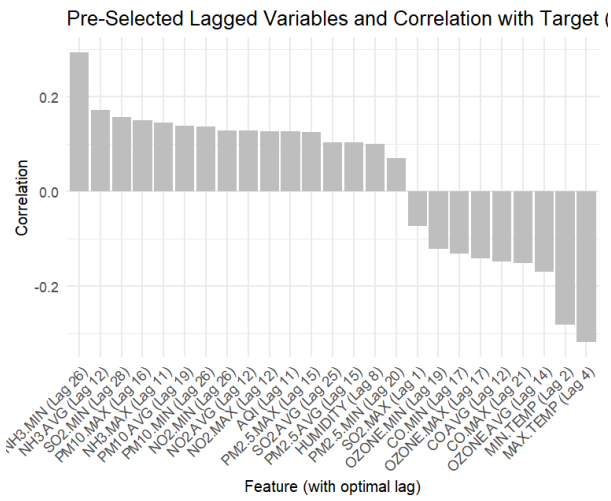


Figure 10: Bar plot of pre-selected lagged features with their correlation to the target (LogReg).

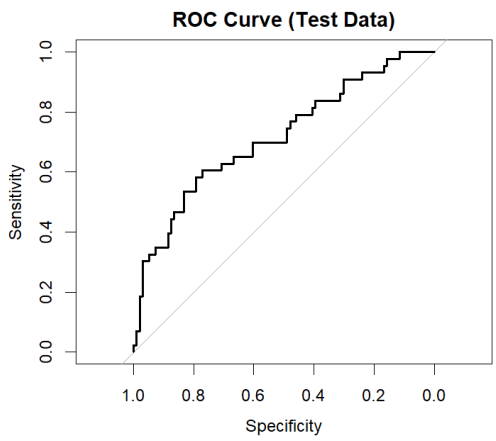
F Appendix F: Logistic Regression Analysis Results

```
Call:
glm(formula = final_formula, family = binomial, data = train_data)

Coefficients:
                 Estimate Std. Error z value Pr(>|z|)
(Intercept)      1.720644   0.826354   2.082  0.03732 *
MAX.TEMP_lag4   -0.140172   0.042954  -3.263  0.00110 **
SO2.AVG_lag25    0.037321   0.014474   2.578  0.00992 **
CO.MAX_lag21    -0.012667   0.004194  -3.020  0.00253 **
NO2.MAX_lag12    0.003150   0.001547   2.036  0.04177 *
NH3.MIN_lag26    0.062903   0.027846   2.259  0.02388 *
SO2.MIN_lag28    0.074364   0.037692   1.973  0.04850 *
PM2.5.MIN_lag20  0.004553   0.002426   1.876  0.06060 .
CO.MIN_lag17    -0.022827   0.010834  -2.107  0.03512 *
MIN.TEMP_lag2    0.086472   0.041001   2.109  0.03494 *
SO2.MAX_lag1    -0.009558   0.005804  -1.647  0.09959 .
PM10.MIN_lag26   0.003785   0.002727   1.388  0.16522
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 710.09  on 555  degrees of freedom
Residual deviance: 597.33  on 544  degrees of freedom
AIC: 621.33
```




```
> # Accuracy
> accuracy <- mean(predicted_classes == test_data$High_Admissions)
> cat("Accuracy (Test Data):", accuracy, "\n")
Accuracy (Test Data): 0.7410072
> # F1 Score
> f1_score <- calculate_f1(confusion_matrix)
> cat("F1 Score (Test Data):", f1_score, "\n")
F1 Score (Test Data): 0.5135135
> # ROC-AUC
> roc_curve <- roc(test_data$High_Admissions, predicted_probabilities)
Setting levels: control = 0, case = 1
Setting direction: controls < cases
> auc_value <- auc(roc_curve)
> cat("ROC-AUC (Test Data):", auc_value, "\n")
ROC-AUC (Test Data): 0.7146318
```

Cross-Validation Metrics by Fold

Fold	Accuracy	AUC	F1_Score
1	0.698	0.716	0.488
2	0.777	0.769	0.608
3	0.727	0.765	0.513
4	0.691	0.679	0.427
5	0.719	0.756	0.494
Mean	0.722	0.737	0.506

Figure 11: (Top left) Summary of Logistic Regression Output; (Top right) ROC Curve for Test Data; (Bottom left) Performance Metrics for Test Data; (Bottom right) Cross-Validation Metrics by Fold.

G Appendix G: ANOVA Support (AQI Distribution)

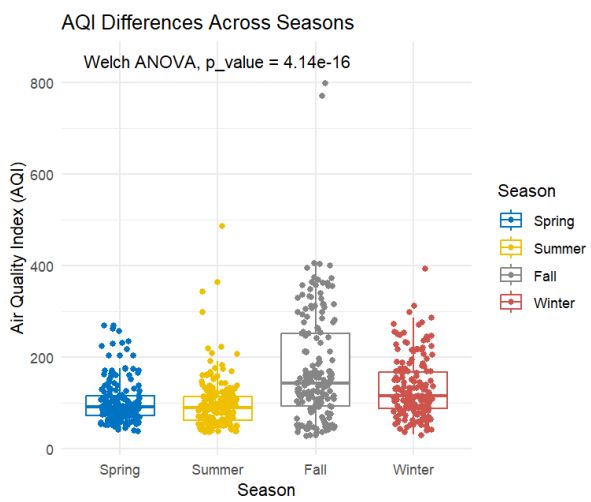


Figure 12: Box plots of AQI distributions across seasons.